Unit 1: Behavior of perfect gases and kinetic theory of gases – 9 hours

Equation of state of a perfect gas, work done in compressing a gas. Kinetic theory of gases – assumptions, the concept of pressure. Kinetic interpretation of temperature; rms speed of gas molecules; degrees of freedom, the law of equipartition of energy (statement only) and application to specific heat capacities of gases; the concept of the mean free path, Avogadro's number.

Equation of state of a perfect gas

A perfect gas (or ideal gas) can be defined as a gaseous mass which obeys the following relation

$$\frac{pv}{T} = R \qquad \text{…………(1)}$$

where p is the pressure (in dynes per square centimetre) exerted by a given mass of the gas occupying a volume v at absolute temperature T. R is a constant which depends on the mass of the gas and is independent of its pressure, volume and temperature. The mathematical relation connecting the three thermodynamic parameters p, v, and T of a substance is called its equation of state.

Deriving the Ideal Gas Equation

Let us assume that the pressure of a gas is 'p,' and the volume of the gas is 'v.' Also, let the temperature be 'T,' R is the universal gas constant, and n is the number of moles of gas. Hence, according to Boyle's Law, if the values of n and T are kept constant, then the volume is inversely proportional to the pressure that is exerted by the gas. This can be represented as:

$V \propto 1/P$

According to Charle's Law, if the values of p and n are kept constant, then the volume of the gas is directly proportional to the temperature. This can be represented as:

$V \propto T$

According to Avogadro's Law, if both P and T are kept constant, then the volume of the gas would be directly proportional to the number of moles of the gas. This can be represented by

$V \propto n$

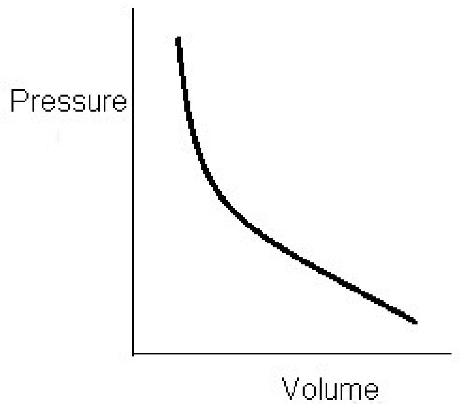
If we combine all the three equations, then

$V \propto nT$ or $PV = nRT$

Work done in compressing a gas

Isothermal Processes

- Isothermal: - Iso means same and thermal related to temperature. In Isothermal process the temperature remains constant throughout while all other variables change.
- Temperature is constant throughout.
- For an ideal gas
    - PV = nRT where
    - n = no. of moles (constant), R = universal gas constant, T =constant for isothermal process.
- This implies PV = constant
- Pressure and volume are inversely proportional to each other.
- Graphically if we plot pressure and volume



- We will get decreasing curve because if we increase pressure volume decreases and vice versa.
- This curve is known as Isothermal Curve.

<u>Isothermal Expansion of an Ideal gas</u>

- It can be described as amount of work done during isothermal expansion of an ideal gas under constant temperature.
- Initially ideal gas is at Pressure P_1 and Volume V_1 .
- At constant temperature the gas will expand from pressure P_1 to P_2 and volume changes from V_1 to V_2. So the final state (P_2,V_2).
- These all expansions are Quasistatic processes.
- Consider any intermediate stage ,
    - Pressure is P and volume is $V_1 + \Delta V$ where ΔV increase in volume.
    - $\Delta W = P\ \Delta V$ where ΔW = small work done
    - By solving and doing calculation the above equation the work done for an ideal gas the work done will be given as :-

$$\mathbf{W = RT\ \ln V_2/V_1}$$

- In an isothermal expansion there is no change in the internal energy of an isothermal process. As there is no change in temperature as a result there is no change in internal energy.
- From First law of thermodynamics
- $\Delta Q = \Delta U + \Delta W$ (putting $\Delta U=0$ as $\Delta T=0$)
- $\Delta Q = \Delta W$.
- This means the amount of heat supplied to the gas is equal to the amount of work done by the gas on the surroundings.

Isothermal expansion and contraction

- The expression of isothermal process for the work done is:
- $W = RT \ln V_2/V_1$
- In isothermal expansion the work is done by the gas whereas in isothermal contraction the work is done on the gas.
- Case 1: Isothermal expansion: $V_2 > V_1$
- This implies $W > 0$, work is positive which means work is done by the gas that is heat is absorbed by the gas.
- Case 2: Isothermal contraction: $V_2 < V_1$
- This implies $W < 0$, work is negative which means work is done on the gas that is heat is released.

Kinetic theory of gases – assumptions

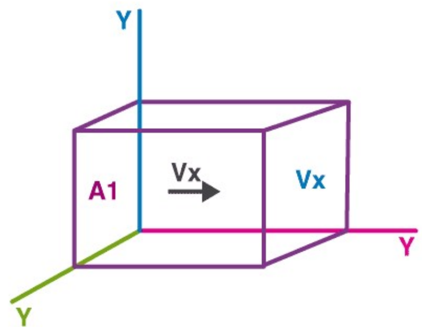
1. Every gas consists of extremely small particles known as molecules. The molecules of a given gas are all identical but are different from those of another gas.
2. The molecules of a gas are identical spherical, rigid and perfectly elastic point masses.
3. Their molecular size is negligible in comparison to intermolecular distance (10^{-9} m).
4. The speed of gas molecules lies between zero and infinity (very high speed).
5. The distance covered by the molecules between two successive collisions is known as free path and mean of all free path is known as mean free path.
6. The number of collision per unit volume in a gas remains constant.
7. No attractive or repulsive force acts between gas molecules.
8. Gravitational to extremely attraction among the molecules is ineffective due small masses and very high speed of molecules.

The concept of pressure

The **pressure** of a gas is a measure of the linear momentum of the molecules. As the gas molecules collide with the walls of a container, the molecules impart momentum to the walls, producing a force that can be measured. The force divided by the area is defined to be the **pressure**.

To find pressure of a gas

Imagine an ideal gas contained in a cubical container. Let one corner of the cube be the origin O, and let the x, y, z-axes along the edges. Let A_1 and A_2 be the parallel faces perpendicular to the x-axis. Consider a molecule moving with velocity v in the container. The components of velocity along the axes are v_x, v_y, and v_z. Now, when the molecule is colliding with face A_1, the x component of velocity is reversed while the y and z component of velocity remains unchanged (as per our assumption that the collisions are elastic).



The change in momentum of the molecule is:
$\Delta P = (-mv_x)-(mv_x) = -2mv_x$ ……….. (1)
Since the momentum remains conserved, the change in momentum of the wall is 2mvx

After the collision, the molecule travels towards the face A2 with x component of the velocity equal to $-v_x$.
Distance travelled from A1 to A2 = L
Therefore, time = L/v_x
After collision with A2, the molecule again travels to A1. Thus, the time between two collisions is $2L/v_x$
Therefore, the number of collisions of the molecule per unit time:

$n = v_x/2L$ ………………… (2)
Using equations 1 and 2,

The momentum imparted per unit time to the wall by the molecule:

$\Delta F = n\Delta p$
$= (m/L)v_x^2$
Therefore, total force on the wall A1 due to all the molecules is

$F = \Sigma(m/L)\, v_x^2$
$= (m/L)\Sigma\, v_x^2$
Now, $\Sigma\, v_x^2 = \Sigma\, v_y^2 = \Sigma\, v_z^2$ (symmetry)
$= 1/3\ \Sigma v^2$
Therefore, $F = 1/3.(m/L)\Sigma v^2$
Pressure is force per unit area so that

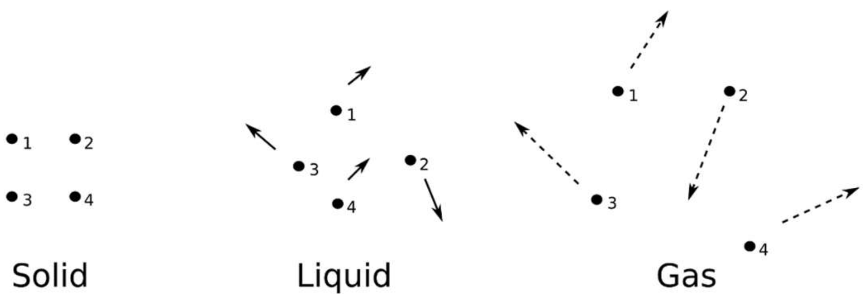
$P = F/L^2$
$= 1/3\ (M/L^3) \sum v^2/N$
$= 1/3\ \rho \sum v^2/N$
Where,

M= total mass of gas

ρ = density of the gas
Now, $\sum v^2/ N$ is written as v^2 and is called the mean square speed.
$P = 1/3\rho v^2$

Kinetic Interpretation of Temperature

The molecules in solids are bonded together such that the molecules cannot move around freely, but they do vibrate, and these vibrations are random, particle exerts a force on each other, and thus they attain kinetic energy due to motion and potential energy due to the interaction.



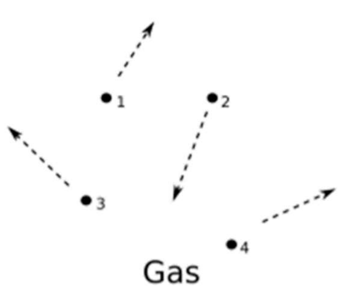
Molecular arrangements in Solids, Liquids and Gases
It's the same case with liquids and gases instead of vibrating they move randomly, now the speed at which they move increases with temperature, cooling down will make them move slowly, therefore,

The temperature of a body is the measure of the average kinetic energy of a body.

We should note that the temperature of a body always depends upon its average kinetic energy and since the average kinetic energy can have a minimum possible value of zero, therefore an object cannot be cooled below a certain minimum value, this value is known as **absolute zero.** This is the lowest possible temperature in our universe and no object could be cooled to this temperature, it is equivalent to -273 degree Celsius or 0 Kelvin. These two scales of temperature can be converted with the following expression,
Temperature in Kelvin = Temperature in degree Celsius + 273

As absolute zero is equal to 0 in the Kelvin scale, it is also known as the absolute scale.

In a gas the particles are always in a state of random motion, all the particles move at different speed constantly colliding and changing their speed and direction,



As the particles collide and change velocity it is not practical to measure each velocity, and as there are particles moving in one direction equal to the particles moving in the opposite direction they cancel out and the average velocity will be equal to zero, hence there is an alternate way to determine their average velocity.

RMS Speed of Gas Molecules

RMS or Root Mean Square method, squaring the velocity, and then taking a square root we can overcome the "direction" component of velocity and still get the average velocity. Since we have removed the direction component, it is no longer average velocity, rather we can call it average speed, and the equation for RMS speed is given as follows,

$v_{rms} = \sqrt{3RT/M}$

Where v_{rms} is the root-mean-square velocity, M is the molar mass of the gas (Kg/mole), R is the molar gas constant, and T is the temperature in Kelvin.

We can see that v_{rms} reduce as the temperature approaches absolute zero.

The rms speed takes both molecular weight and temperature into account, which are the factors that directly affect the kinetic energy of a gas.

Kinetic interpretation of temperature (Derivation)

According to kinetic theory of gases pressure of an ideal gas is given by

$$P = \frac{1}{3}\rho v^2 = \frac{1}{3}\frac{M}{V}v^2$$

where, v is the mean of the square speed.

$$\Rightarrow \quad PV = \frac{1}{3}Mv^2$$

Now, average translational kinetic energy of the molecule is given by,

$$\Rightarrow \quad \frac{1}{2}Mv^2 = \frac{3}{2}PV = \frac{3}{2}nRT$$

$$\Rightarrow \quad \frac{1}{2}Mv^2 = \frac{3}{2}nNkT$$

$$\Rightarrow \quad \frac{1}{2}\left(\frac{M}{nN}\right)v^2 = \frac{3}{2}kT$$

$$\text{or} \quad \frac{1}{2}mv^2 = \frac{3}{2}kT$$

T: absolute temperature.

This is the relation between average kinetic energy of a molecule of gas and absolute temperature of the gas.

According to this relation average kinetic energy of a molecule of an ideal gas is proportional to its absolute temperature; is indeppendent of the pressure, volume or the nature of the ideal gas. This is called kinetic interpretation of temperature.

Degree of Freedom

The degree of freedom for a dynamic system is the number of directions in which it can move freely or the number of coordinates required to describe completely the position and configuration of the system.

It is denoted by for N.

Degree of freedom of a system is given by

f or N = 3A – R

where A = number of particles in the system and R = number of independent relations.

Degree of Freedom

1. For monoatomic gas = 3
2. For diatomic gas = 5
3. For non-linear triatomic gas = 6
4. For linear triatomic gas = 7

Specific heat of a gas

(a) At constant volume, $C_V = f/2\ R$
(b) At constant pressure, $c_p = (f/2 + 1)R$
(c) Ratio of specific heats of a gas at constant pressure and at constant volume is given by $\gamma = 1 + 2/f$

Law of equipartition of energy: Statement

According to the law of equipartition of energy, for any dynamic system in thermal equilibrium, the total energy for the system is equally divided among the degree of freedom.

The kinetic energy of a single molecule along the x-axis, the y-axis, and the z-axis is given as

$\frac{1}{2}mv_x^2$, along the x-axis
$\frac{1}{2}mv_y^2$, along the y-axis
$\frac{1}{2}mv_z^2$, along the z-axis

When the gas is at thermal equilibrium, the average kinetic energy is denoted as

$\langle\frac{1}{2}mv_x^2\rangle$,along the x-axis
$\langle\frac{1}{2}mv_y^2\rangle$, along the y-axis
$\langle\frac{1}{2}mv_z^2\rangle$, along the z-axis

Mean Free Path

The average distance travelled by a molecule between two successive collisions is called mean free path (γ).

Mean free path is given by

$\gamma = kT / \sqrt{2} \pi \sigma^2 p$

where σ = diameter of the molecule, p = pressure of the gas, T = temperature and k = Botlzmann's constant.

Mean free path

$\lambda \propto T$ and $\lambda \propto 1/p$

Brownian Motion

The continuous random motion of the particles of microscopic size suspended in air or any liquid, is called Brownian of microscopic motion.

Brownian suspended motion in both is observed with many liquids and gases.

Brownian motion is due to the unequal bombardment of the suspended Particles by the molecules of the surrounding medium.

Avogadro's Law

Avogadro stated that equal volume of all the gases under similar conditions of temperature and pressure contain equal number molecules. This statement is called Avogadro's hypothesis. According Avogadro's law

(i) Avogadro's number The number of molecules present in 1g mole of a gas is defined as Avogadro's number.

$N_A = 6.023 \times 10^{23}$ per gram mole

(ii) At STP or NTP (T = 273 K and p = 1 atm 22.4 L of each gas has 6.023×10^{23} molecules.

(iii) One mole of any gas at STP occupies 22.4 L of volume.